UNIVERSITY OF INFORMATION TECHNOLOGY
FACULTY OF INFORMATION SYSTEMS

Introduction to the course

Principles of Database Systems

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INTRODUCTION

1. Course name: Principles of Database Systems
2. Credits: 4 (3 lectures – 1 lab)
3. Course description
4. Contents
5. Requirements and Expectations
6. Assessment
7. References

Course description

- The course introduces fundamental concepts of computer database and use.
- It provides a study of data models, data description languages, and query facilities including relational algebra and SQL, data normalization, methodologies of database design, techniques for database application development.
- *It also discusses certain other aspects of database systems: recovery, concurrency, security, and integrity.*

Content

1. Introduction
2. Data Modelling
3. Process of Database Design
4. Database and Database Application Design
5. Database Implementation
6. Functional Dependencies
7. Trends in Database

Requirements and Expectations

- Preparation
- Attendance (NOT absent over 2 class sessions)
- Group discussion
- Homework

Assessment

1. Midterm exam: 20%
2. Computer based test: 30%
3. Final exam: 50%
4. Bonus score: Max +5% to [2]
   - Excellent students (Discussion, homework, learning attitude, and other activities in class)

References

1. Ramez Elmasri, and Shamkant B. Navathe, *Fundamentals of Database Systems (7th Ed.)*, Addison-Wesley, 2016.

2. Nguyen Dang Ty, and Do Phuc, *Database*, Vietnam National University – HCMC, 2015.

3. George Tillmann, *Usage-Driven Database Design,* Apress, 2017.

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